

1. Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation?

- Kill processes by name

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ pacman -S psmisc
resolving dependencies...
looking for conflicting packages...

Packages (1) psmisc-23.7-1
Total Download Size: 0.13 MiB
Total Installed Size: 0.64 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
psmisc-23.7-1-x86_64
(1/1) checking keys in keyring
(1/1) checking package integrity
(1/1) loading package files
(1/1) checking for file conflicts
(1/1) checking available disk space
:: Processing package changes...
(1/1) installing psmisc
john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ sleep 500
Terminated          sleep 500
[1]+  Terminated   sleep 500

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MINGW64 ~
$ cd /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ps aux | grep sleep
379    359    379    15832  ptty1    197610 16:42:40 /usr/bin/sleep

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ kill 379

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ps aux | grep sleep

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

- process based on process name Kill

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ pacman -S psmisc
resolving dependencies...
looking for conflicting packages...

Packages (1) psmisc-23.7-1
Total Download Size: 0.13 MiB
Total Installed Size: 0.64 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
psmisc-23.7-1-x86_64
(1/1) checking keys in keyring
(1/1) checking package integrity
(1/1) loading package files
(1/1) checking for file conflicts
(1/1) checking available disk space
:: Processing package changes...
(1/1) installing psmisc
john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ sleep 500
Terminated          sleep 500
[1]+  Terminated   sleep 500

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ killall sleep

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ps aux | grep sleep

john@DESKTOP-JAOTCLF MINGW64 /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

- Kill a single process at a time with the given process ID

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS ~
$ cd /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ sleep 400
Terminated          sleep 400

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |

/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS ~
$ cd /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ps aux | grep sleep
471    436    471    15028  ptty1    197610 10:41:17 /usr/bin/sleep

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ kill 471

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ps aux | grep sleep

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

2. Write a program for process creation using C

●Orphan Process

```
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child exiting\n");
    }
    else {
        sleep(10);
        printf("Parent running\n");
    }
    return 0;
}
```

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano orphan.c
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc orphan.c -o orphan
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./orphan
Parent exiting
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ Child Process
PID : 490
PPID : 1
```

●Zombie Process

```
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child exiting\n");
    }
    else {
        sleep(10);
        printf("Parent running\n");
    }
    return 0;
}
```

```
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano zombie.c
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc zombie.c -o zombie
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./zombie
Child exiting
Parent running
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ps -el
  PID   PPID  PGID   WINPID   TTY      UID    STIME  COMMAND
    711     575    711     11208   pty0     197610 19:40:12 /usr/bin/ps
    575     574    575     17604   pty0     197610 19:22:51 /usr/bin/bash
    574      1    574     12616   ?        197610 19:22:51 /usr/bin/mintty
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

3. Create the process using fork () system call.

●Child Process creation

●Parent process creation

●PPID and PI

```
M /c/Users/john/OneDrive/Desktop/OSY_CD25D005
```

```
GNU nano 8.7
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
```

```
int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed\n");
    }
    else if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
    }
    else {
        // Parent process
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
        wait(NULL);
    }
    return 0;
}
```

```
M /c/Users/john/OneDrive/Desktop/OSY_CD25D005
```

```
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano fork.c
```

```
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc fork.c -o fork
```

```
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./fork
```

```
Parent Process
```

```
PID : 720
```

```
Child PID : 721
```

```
Child Process
```

```
PID : 721
```

```
PPID : 720
```

```
john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```